

AutoPooper 3000 — Sifter Litter Capacity Simulation

2026-10-02 · Sifter `AutoPooper_Sifter_Assy_v6_drumfit` with `Sifter_Honeycomb_Screen_v2_wide` · Drum `AutoPooper_Drum_v2_nostar` (3mm spindle shim)

Summary

The sifter can carry **about 4.1 L (worst case) to 4.75 L (realistic)** of clean litter through the dump turn to the 234° stop. Because the drum is a globe, that is a litter bed about **7 cm deep** at its deepest point. Recommended fill: **3.5–4 L** (a 6.5–7 cm bed), which is about **3.5–4 kg** of clumping clay litter. That matches what store-bought rotating boxes use: the Litter-Robot 3 and 4 both take about 4 L (1 gallon, 8–10 lb).

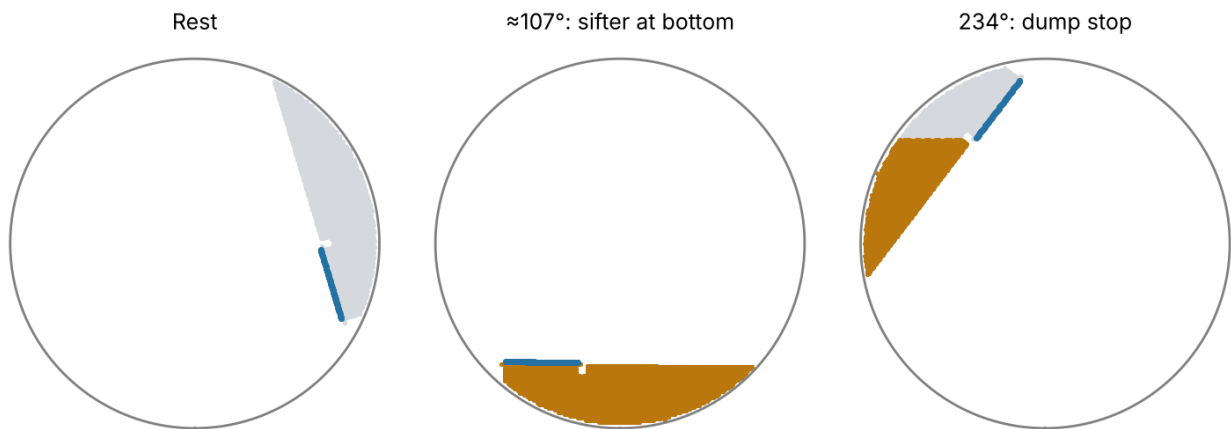
Item	Value
Sifter hollow volume (after screen, frame and dam)	about 6.5 L
Most it holds (at the bottom of the turn)	about 6.4 L
Carried to the 234° dump stop, worst case	4.07 L
Carried to the 234° dump stop, realistic (litter piles at ~32°)	4.75 L
Recommended fill	3.5–4 L (6.5–7 cm bed), about 3.5–4 kg
Gain from stopping at 226° instead of 234°	only 0.1–0.2 L more

What was simulated

Placement. The sifter sits where your mockup sat: about 107° from straight down at rest, flat end toward the front of the drum, slot on the leading edge and the big hollow trailing. Its back was reshaped to follow the drum's inner wall with a 0.5 mm glue gap (`v6_drumfit`), seated 2° nose-up and 0.5° rolled to get the best fit.

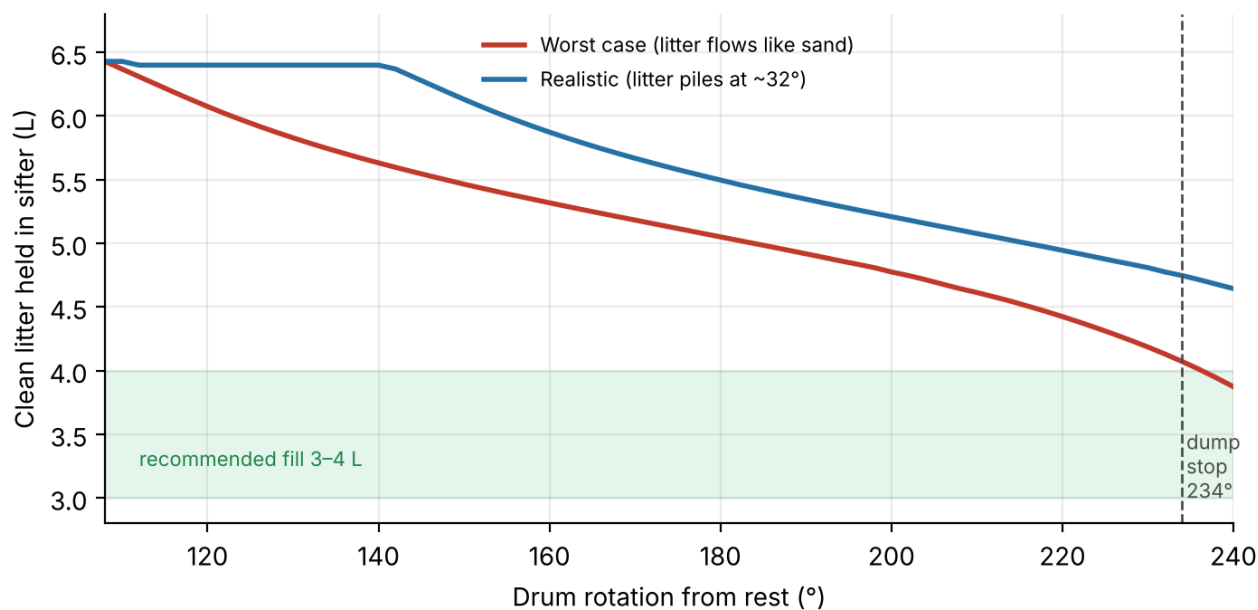
Motion. The drum turns clockwise (seen from the front) from rest to the 234° dump stop. The sifter passes the bottom at about 107°, then rises up the other side, ending about 56° short of the top at the dump stop.

Method. The hollow was split into 3 mm cubes (about 241,000). At every 2° of rotation, litter stays only below the lowest point of the screen opening; anything above that line pours out. Litter that pours out is not counted back in, so the carried amount is the lowest value seen between the bottom pass and the dump stop. Two bounds were run: **worst case**, where litter flows like sand to a flat level, and **realistic**, where it can hold a slope of up to about 32°, as clumping clay does.



Slice through the middle of the sifter, seen from the front. Grey: hollow; blue: screen opening; brown: litter held (worst case).

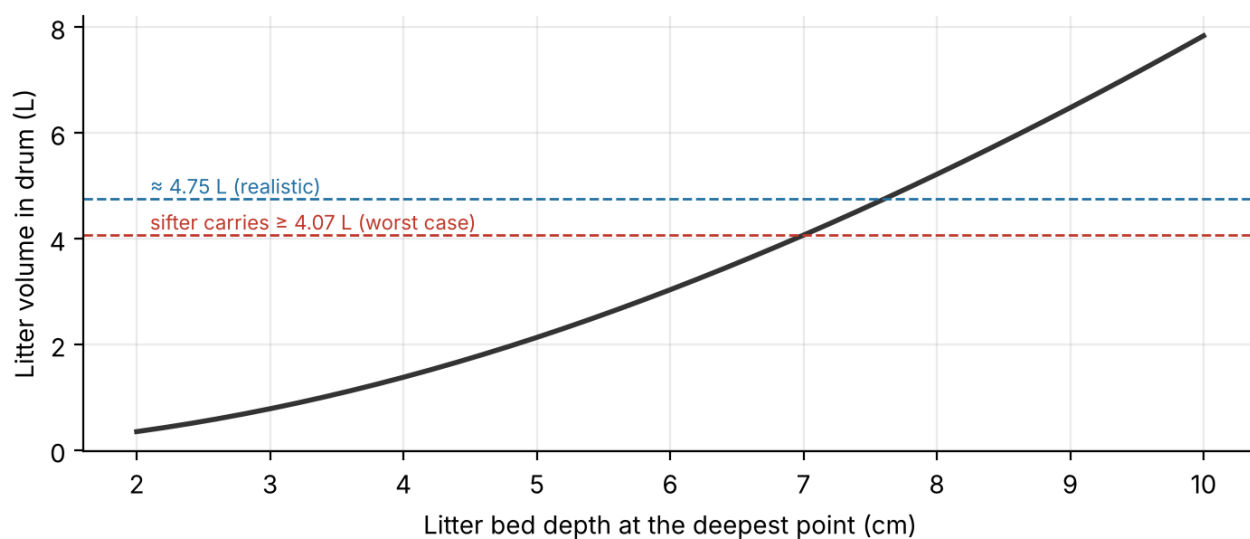
Results



Drum rotation	Worst case (L)	Litter piles at 25° (L)	Litter piles at 32° (L)
110° (at bottom)	6.37	6.43	6.43
130°	5.83	6.43	6.40
160°	5.32	5.73	5.87
190°	4.92	5.25	5.35
220°	4.42	4.84	4.94
226° (your mockup stop)	4.29	4.76	4.86
234° (current dump stop)	4.07	4.63	4.75
240°	3.87	4.53	4.64

Litter bed depth vs. volume

The drum is a globe about 574 mm across inside, so a given bed depth holds much less litter than a straight cylinder would.



Bed depth at deepest point	3 cm	4 cm	5 cm	6 cm	7 cm	8 cm	9 cm
Litter in drum (L)	0.8	1.4	2.1	3.0	4.1	5.2	6.5

How much litter to buy

Clumping clay (bentonite) weighs about 0.9–1.1 kg per litre, so the 3.5–4 L fill is about **3.5–4 kg**. Buy one **10 kg bag** (or about 10 L if the bag is sold by volume, as many are in the Philippines). That covers the first fill plus a few weeks of top-ups. Measure the first fill with a jug rather than by weight, because density varies by brand. Then mark a fill line inside the drum, as the Litter-Robot does, so it never gets overfilled.

Limits of this simulation

- It is a slow-motion model. It doesn't model grains bouncing or sliding while the drum moves, or the shakes at the dump stop.
- **Filling isn't modelled.** With a 7 cm bed the slot only dips into the litter between about 60° and 95° of rotation, as its leading edge sweeps through. Whether the sifter fills completely depends on litter piling up against it. If it under-fills in testing, a pause or a few small rocks back and forth around 80–100° in the firmware should help.
- The internal dam mainly helps against sloshing during real motion, so its benefit doesn't show up in this model.
- Bed volumes use the drum's measured inner shape, averaged around the drum.
- Check it for real: run one cycle with a measured amount of clean litter (no clumps) and weigh what ends up in the receptacle.